FUTURE HEALTH | A GLOBAL INITIATIVE BY ABU DHABI | H2S HACK2SKILL

# Future Health Digital Innovation Challenge

From signals to solutions.

Co-creator Microsoft

Co-creator inception42

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Team Name: Design Master Solutions

Team Size: 1

Problem Statement: [Population Health] Closing the Screening-to-Care Loop.

Solution: Loop Closer

Team members*
[Name, country, role, organization, age confirmation]

| S.No | Name | Country | Role | Organization | Age confirmation |
|---|---|---|---|---|---|
| 1 | Alan De Bortolo | Brazil | Founder, lead developer | Design Master Solutions LLC (USA) | Confirmed: 18 or older |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |

## Team Capability Summary*

*Maximum 1,000 characters*

Design Master Solutions LLC is a Massachusetts software company founded in 2019. It designs, codes, hosts and supports its own products, with web, mobile and backend work done in house.

The founder also builds All4Clinic, a clinic management system in Brazil used by 24 paying clinics. Its appointment confirmation reads free-text patient replies and opens a task for whoever does not answer, the same pattern this challenge needs. No All4Clinic code or data is used here.

Gap we state openly: the team has no clinician. Before any pilot, the clinical lead of the screening program must own the pathway, the service levels and the patient wording.

## Architecture or workflow diagram*

**Case pathway.** One accountable owner and one due date for every step.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Result issued | Patient notified | Understanding confirmed | Confirmatory test | Referral sent | Referral accepted | Appointment attended | Treatment or surveillance plan | Closed with evidence |

**INPUT**
HL7 v2 ORU^R01 or FHIR R4 from the lab or program. One case per accession number. Built and tested on sample messages

**CASE ENGINE**
service level by urgency, owner per step; a clock alerts the owner of each late step or missed visit and escalates urgent cases

**HUMAN CONTROL**
Coordinator worklist shows only cases that need a person. Only a clinician can close, with evidence and outcome

**PATIENT**
Plain message in Arabic or English, never a diagnosis. The reply moves the case: 1 understood or attended, 2 nurse, 3 new date

**CONSULTATION**
Patient-to-clinician and clinician-to-clinician questions on the case, with their own service level

**MEASURES**
Notified, understood, referral completion, days to specialist, unresolved, why cases stalled

Built as pure functions with no I/O (public/loop.mjs), so the same logic runs in the browser and in the tests. No AI decides anything clinical in this prototype.

Target users and intended outcome*

**Screening coordinators and care navigators:** start the day with the short list of cases that need a person, not a spreadsheet of every result.

**Primary care doctors and specialists:** see who owns the next step across organizations, accept or reject a referral, and answer questions on the case.

**Patients:** get a clear next step in their language, confirm they understood, and reach a nurse with one reply.

**Program leaders:** see where cases wait and why, with documented closure for each actionable result.

**Intended outcome:** every actionable screening result is communicated, confirmed and converted into timely care, with a named owner and documented closure. No result stalls without somebody seeing it.

## How will success be measured?*

Prototype scenario on the same 60 simulated cases: a stall stays silent, or the worklist surfaces it and coordinators resolve 80% of exceptions two days after the breach. Model assumptions, not a measured outcome.

| Measure | Silent | Worklist |
|---|---|---|
| Closed with evidence | 19 | 29 |
| Stalled, nobody acting | 23 | 7 |
| Patient notified | 93% | 98% |
| Understanding confirmed | 89% | 95% |
| Referrals completed | 67% | 73% |

**In a pilot, against the program baseline:**

Time to notification, confirmation, consultation and care

Referral and appointment completion

Unresolved actionable results and reasons for drop-off

Share of closures with documented evidence

Patient understanding and confidence, by language

Median days to specialist does not move in this scenario: recovered cases arrive later, so it needs real data.

## Evidence supporting the approach*

*Citations, studies, benchmarks, or prior testing.*

**Failure is common.** Callen et al., J Gen Intern Med 2012;27:1334-48: systematic review of 19 studies; tests not followed up ranged from 6.8% to 62% for laboratory tests and 1.0% to 35.7% for radiology, with missed cancer diagnoses among the impacts.

**Delay matters.** Corley et al., JAMA 2017;317:1631-41: 70,124 patients with a positive FIT; follow-up after 10 months was associated with higher colorectal cancer risk and more advanced stage than follow-up at 8 to 30 days.

**Follow-up needs a system.** Doubeni et al., CA Cancer J Clin 2018;68:199-216: PROSPR systematic review and recommendations on timely follow-up of positive cancer screening results. Selby et al., J Med Screen 2021;28:51-53: international survey of interventions that ensure follow-up of positive FIT.

**Prior testing.** 39 automated tests pass. FHIR output checked with the official HL7 validator and round-tripped on a public FHIR R4 server (17 resources created, read back, no duplicates on resend). No clinical testing yet.

## Data sources and legal right to use them*

*List source, type, owner, sensitivity, permissions, and whether personal data is used.*

| Source | Type | Owner | Personal data | Permission |
|---|---|---|---|---|
| Simulated cohort (60 cases, fixed seed) | Synthetic | Design Master Solutions | No | Own work, MIT license |
| Staff names in the demo | Fictional | Design Master Solutions | No | Invented for the demo |
| Patient messages (EN, AR) | Text written for the prototype | Design Master Solutions | No | Needs clinical and language review before use |
| Pilot: screening results and referrals | Health data, high sensitivity | Screening program and providers | Yes | Not used. Requires program agreement, health authority approval and patient consent basis |

## Risks and safeguards*

*Privacy; security; bias; explainability; misuse; human oversight; failure modes.*

**Privacy and security:** no patient data in the prototype, and the public API refuses any message not marked as test data. A pilot needs UAE hosting, role-based access, an audit log and encryption.

**Bias:** rule-based, no model scoring patients. We report measures by language and program to catch unequal drop-off.

**Explainability:** every flag comes from a visible rule: owner missing, due date passed, referral rejected.

**Misuse:** patient text always says it is not a diagnosis. Wording is reviewed by the program.

**Human oversight:** urgent cases with any flag go to the top. Only a clinician closes, and only with evidence.

**Failure modes:** built: silent-feed alarm, reconciliation against the lab list, corrected results flag the open case, unknown or unreadable results go to a person. In a pilot, unsent messages must surface as exceptions. Service levels are illustrative until the program sets them.

## Hardware requirements

*Required if wearable, sensor, connected device, or other hardware is there. Include sourcing and safety.*

Not applicable. Loop Closer is software only and needs no wearable, sensor or connected device.

## Path to pilot and scale*

*Stakeholders, integration, regulatory needs, timeline assumptions, and sustainability.*

**Stakeholders:** Department of Health Abu Dhabi, screening program leads, laboratories, primary care, specialist services, a named clinical lead.

**Integration:** built and tested on sample data: results in by HL7 v2 ORU^R01 or FHIR R4; case state out as an idempotent FHIR R4 transaction (Task, ServiceRequest, Communication, Encounter, CarePlan, Provenance). Next: a production lab interface and Malaffi, the Abu Dhabi health information exchange. Scale: over 10,000 messages per second on one laptop core (docs/bench.txt).

**Regulatory:** UAE Federal Law No. 2 of 2019 on ICT in health fields (health data stays in the UAE unless an exception applies; long retention). Formal legal review before any pilot.

**Timeline assumption:** 1 to 2 months to agree pathway and service levels with one program; 3 months of integration and security review; 3 to 6 months of shadow pilot on one screening program, then a measured comparison with its baseline.

**Sustainability:** annual subscription per program, justified by measured closure. Pricing to be validated in the pilot.

## Pre-existing IP and third-party materials*

*Identify ownership, licenses, open-source components, and restrictions.*

**Ownership:** all code was written for this challenge from 30 October 2026, in a new repository owned by Design Master Solutions LLC, MIT license: github.com/DesignMaster-Solutions/loop-closer.

**Pre-existing IP:** none. No code or data from All4Clinic or other company products.

**Open-source and third-party:** no runtime dependencies. Node.js test runner for tests, HL7 FHIR Validator and the public HAPI test server to check output, Wrangler to deploy, Cloudflare Workers to host the demo.

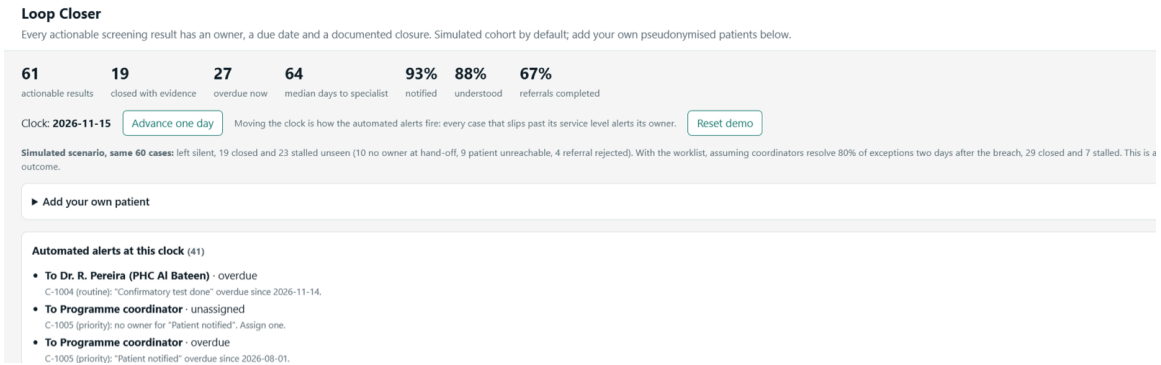
**AI assistance:** built with an AI coding assistant, reviewed and tested by the team.

**Restrictions:** none known in the prototype. Clinical wording and pilot data would belong to the program.

## Conflict of interest declaration*

*Disclose relationships with organizers, partners, or judges.*

None. No member of our team has a conflict of interest to declare.




**Review round, 5 November 2026:** your own patients and results; a demo clock whose automated alerts reach the owner of each late step; missed visits flagged with a rebooking deadline; patient replies 1, 2 or 3 that move the case; closure with signer, evidence and outcome.

**Live demo:** loop-closer.designmaster.workers.dev

**Repository:** github.com/DesignMaster-Solutions/loop-closer

**Limitations:** simulated cohort by default, or your own pseudonymised patients saved only in the browser; service levels and scenario rates are assumptions; Arabic text needs review; FHIR and HL7 v2 tested on sample data only, no production connection; demo changes stay in the browser.

Additional Slide

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# Thank you

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